

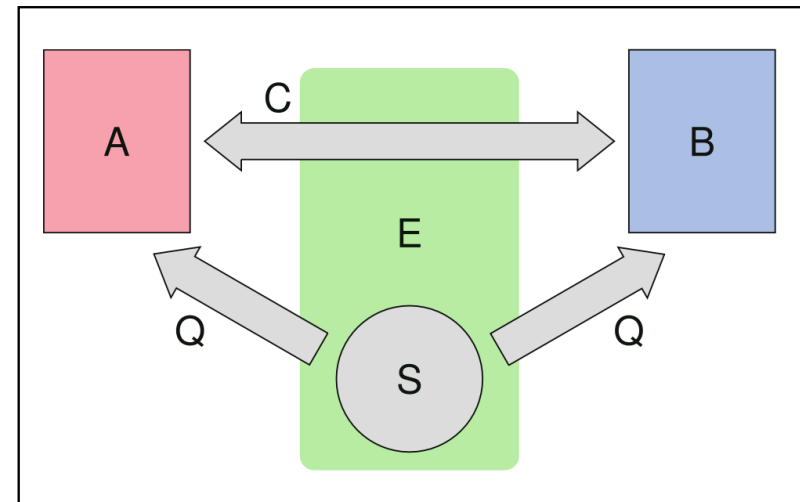
Protocol Setup and Security Features



1. Alice and Bob use:
 1. A **quantum channel** from the source to receive entangled states.
 2. An **authenticated classical channel** to communicate (Eve can listen, but not alter messages).
2. Security Analysis:
 1. Easier compared to prepare-and-send protocols.
 2. No direct quantum channel needed between Alice and Bob.

The setting of an entanglement-based protocol:

1. Alice (A) and Bob (B) receive entangled states via quantum channels (Q).
2. Eve (E) can listen to the classical channel (C) but cannot interfere.





The Ekert91 Protocol

1. Developed by Arthur Ekert in 1991 using **maximally entangled qubit pairs**.
2. The source distributes **maximally entangled states** to Alice and Bob.
3. Example state used in Ekert91:

$$|\Psi^-\rangle_{AB} = \frac{1}{\sqrt{2}}(|01\rangle_{AB} - |10\rangle_{AB})$$

4. Alice and Bob perform measurements on their qubits:
 1. Randomly choose observables A_i and B_i from specific sets.
 2. Observables lie in the x-z plane of the Bloch sphere.

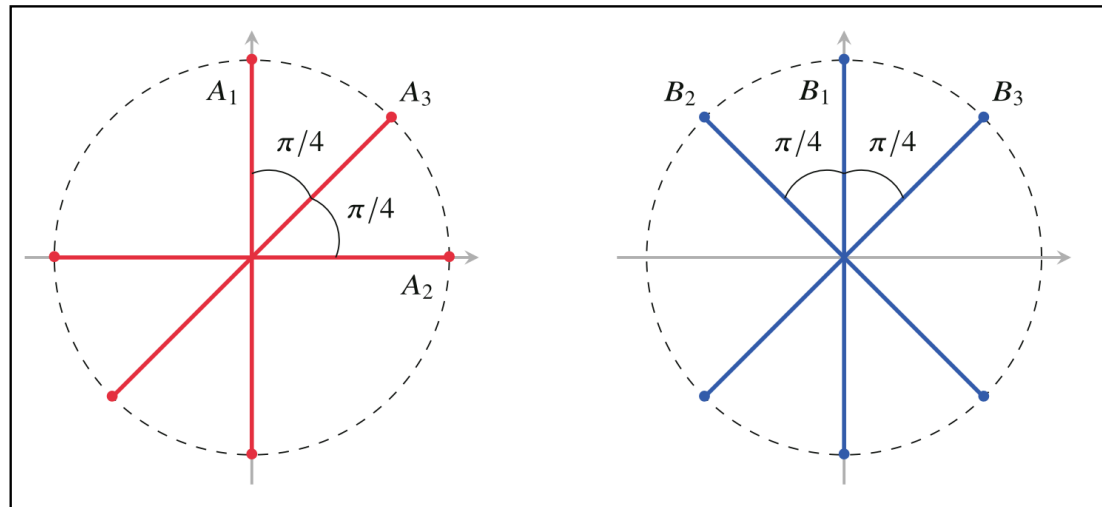
Measurement Settings in Ekert91

1. Alice and Bob randomly choose measurements from:

1. **Alice:** $A_1 = Z$, $A_2 = X$, $A_3 = \frac{1}{\sqrt{2}}(Z + X)$

2. **Bob:** $B_1 = Z$, $B_2 = \frac{1}{\sqrt{2}}(Z - X)$, $B_3 = \frac{1}{\sqrt{2}}(Z + X)$

2. These observables correspond to spin components in the $x - z$ plane.



Measurement directions for the Ekert protocol.

For Alice - left. For Bob - right.

Sifting and Security Steps



1. Alice and Bob share many entangled pairs.
2. Each randomly chooses one measurement from their respective sets.
3. They publicly announce the measurement bases (not results).
4. When bases match: (A_1, B_1) and (A_3, B_3) produce anti-correlated outcomes → **sifted key**.
5. Other basis combinations $(A_1B_3, A_1B_2, A_2B_3, A_2B_2)$ → used for **security check**.



The CHSH Inequality

1. Four classical random variables: A_1, A_2, B_3, B_2 with
Realizations: $+1, -1$

2. $A_1(B_3 + B_2) + A_2(B_3 - B_2) = \pm 2$

3. $\left| \langle A_1(B_3 + B_2) + A_2(B_3 - B_2) \rangle \right| \leq 2$

4. $S := \left| \langle A_1 B_3 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_3 \rangle - \langle A_2 B_2 \rangle \right| \leq 2$

1. **Quantum observables:** A_1, A_2, B_3, B_2

2. **Expectation value:** $\langle A_i B_j \rangle = \text{Tr}(A_i \otimes B_j \rho)$

3. Example for Ekert protocol:

1. $A_1 = Z, \quad B_3 = \frac{1}{\sqrt{2}}(Z + X), \quad \rho = |\Psi^-\rangle\langle\Psi^-|$

2. $\langle A_1 B_3 \rangle = \langle \Psi^- | Z \otimes \frac{1}{\sqrt{2}}(Z + X) | \Psi^- \rangle = -\frac{1}{\sqrt{2}}$

1. $\langle A_i B_j \rangle = \frac{1}{N} \sum_{\nu} A_i^{\nu} B_j^{\nu}$

2. A_i^{ν} = realization of r.v. A_i

1. **Calculate CHSH value S :**

2. $S = |\langle A_1 B_3 \rangle + \langle A_1 B_2 \rangle + \langle A_2 B_3 \rangle - \langle A_2 B_2 \rangle| = 2\sqrt{2}$

3. Classical: $S \leq 2$



CHSH Inequality

Eq1 QKD Protocol. $|\psi^-\rangle_{AB} = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$

$$A_1 = Z, A_2 = X, A_3 = \frac{1}{\sqrt{2}} (Z + X)$$

$$B_1 = Z, B_2 = \frac{1}{\sqrt{2}} (Z - X), B_3 = \frac{1}{\sqrt{2}} (Z + X)$$

Find $\langle A_1 B_3 \rangle = \langle \psi^- | Z \otimes \frac{Z+X}{\sqrt{2}} | \psi^- \rangle$

$$\hookrightarrow A_1 \otimes B_3 = \frac{Z \otimes Z + Z \otimes X}{\sqrt{2}}$$

$$(Z \otimes Z) |01\rangle = -|01\rangle$$

$$(Z \otimes Z) |10\rangle = -|10\rangle$$

$$\Rightarrow (Z \otimes Z) |\psi^-\rangle = -|\psi^-\rangle$$

$$\langle Z \otimes Z \rangle = \langle \psi^- | (-1) | \psi^- \rangle = -1$$

$$\begin{aligned} Z|0\rangle &= |0\rangle \\ Z|1\rangle &= -|1\rangle \end{aligned}$$

$$\begin{aligned} X|0\rangle &= |1\rangle \\ X|1\rangle &= |0\rangle \end{aligned}$$

★ Now

$$Z \otimes X$$

$$(Z \otimes X) |01\rangle = |00\rangle$$

$$(Z \otimes X) |10\rangle = -|11\rangle$$

$$(Z \otimes X) |\psi^-\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$\langle Z \otimes X \rangle = \langle \psi^- | \phi^+ \rangle = 0$$

$$\begin{aligned} \Rightarrow \langle A_1 B_3 \rangle &= \frac{1}{\sqrt{2}} (\langle Z \otimes Z \rangle + \langle Z \otimes X \rangle) \\ &= \frac{1}{\sqrt{2}} (-1 + 0) = -\frac{1}{\sqrt{2}} \end{aligned}$$



CHSH Inequality and Eve Detection

1. CHSH inequality:

$$S := |A_1B_3 + A_1B_2 + A_2B_3 - A_2B_2| \leq 2 \quad (\text{classical})$$

2. Quantum prediction using entangled states:

$$S = 2\sqrt{2} \quad (\text{violates classical bound})$$

3. If $S > 2$, Alice and Bob share a **genuinely entangled state**.

4. If $S \leq 2$, no entanglement $\rightarrow$ possible eavesdropping $\rightarrow$ abort.

5. Any violation confirms security (even partial $\rightarrow$ key can still be distilled).



The Ekert protocol

1. Alice and Bob distribute a number of $|\Psi^-\rangle_{AB}$ states between them.
2. For each state, Alice and Bob randomly choose a measurement from the sets $\{A_i\}$ and $\{B_i\}$.
3. Alice and Bob announce the bases they chose for each measurement. The results of the pairs (A_1, B_1) and (A_3, B_3) form the sifted key.
4. The results for the pairs (A_1, B_3) , (A_1, B_2) , (A_2, B_3) and (A_2, B_2) are used to check the CHSH inequality.
5. If CHSH test passed:
 - (i) Proceed with **error correction**.
 - (ii) Apply **privacy amplification** to eliminate Eve's potential information.
 - (iii) Result: a **shared secret key**.
6. Entanglement guarantees security even if source is untrusted.